Not All Policy Ambiguity Is Equal: Specification-Layer Ambiguity Poses Greater Safety Risks for LLM Agents

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Abstract

Large Language Model (LLM) agents increasingly execute real-world tasks governed by natural-language policies, yet these policies inevitably contain ambiguity. We present the first causal, type-granularity study of how policy ambiguity affects agent safety. Using a matched-pair design over 300 clause pairs spanning six ambiguity types, we evaluate five LLM agents (GPT-5.4, GPT-4.1, Claude Sonnet 4.6, Qwen3-32B, Qwen3-8B) in 3,000 agentic episodes on the τ^2-bench framework. Our core finding is *Differential Nocuity*: while all ambiguity types cause massive safety violation increases ($+35.8$ pp overall, McNemar OR $= 42.1$, $p < 0.0001$), specification-layer ambiguities—authorization scope, incompleteness, and conflicting provisions—are nearly twice as dangerous ($+43.6$ pp) as linguistic-layer ones ($+26.2$ pp; gap $= 17.4$ pp, $p < 0.0001$, $d = 0.68$). A perturbation control confirms the effect is ambiguity-specific, not general prompt brittleness (perturbation $\Delta = 3.3$ pp vs. ambiguity $\Delta = 36.5$ pp). The type hierarchy is architecture-universal: five models across three families maintain the same ranking (Model$\times$Type interaction $p = 0.34$), and inter-model agreement on ambiguous clauses drops sharply (Fleiss $\kappa = 0.33$) compared to unambiguous ones ($\kappa = 0.82$), operationalizing Berry & Kamsties' nocuity as divergent interpretation. Error analysis reveals mechanistically distinct failure modes: specification-layer violations stem from unauthorized assumptions and scope escalation, while linguistic-layer violations arise from misparsing and surface adoption—with coreferential ambiguity uniquely triggering conservative avoidance behavior that partially explains its low violation rate. We release all clause pairs, evaluation code, and annotated traces.

1 Introduction

A customer returns a customized laptop bag within the 30-day window. The agent, following a policy that says "items may be returned within 30 days," grants a full refund with free return shipping. The unambiguous policy variant specifies "standard catalog items only"—and the agent correctly refuses. One word of scopal ambiguity costs $200.

LLM agents are increasingly deployed to handle customer service, financial operations, and enterprise workflows, where their behavior is governed by natural-language policy documents [Yao et al., 2023, Schick et al., 2024]. These documents—return policies, access-control rules, operational guidelines—are written by humans and inevitably contain ambiguity [Berry and Kamsties, 2003, Massey et al., 2014]. In human organizations, shared context, institutional norms, and the ability to ask questions render most ambiguity *innocuous* [Berry and Kamsties, 2003]. LLM agents lack these disambiguation channels.

Prior work has catalogued ambiguity types in policy documents [Massey et al., 2014, Li et al., 2024] and shown that models disagree on ambiguous interpretations, but no study has measured whether—and how much—this disagreement translates into safety violations. Separately, agent safety benchmarks [Ruan et al., 2024, Yuan et al., 2024] evaluate diverse failure modes but do not isolate ambiguity as a causal factor.

We close this gap with the first causal, type-granularity study of policy ambiguity's impact on agent safety. Our matched-pair design pairs each ambiguous clause with a minimally edited unambiguous variant,

isolating the effect of ambiguity from task difficulty. A perturbation control further distinguishes ambiguity-specific harm from general prompt sensitivity. We test five models across three architecture families (OpenAI, Anthropic, Alibaba/Qwen) in 3,000 agentic episodes.

Our findings introduce **Differential Nocuity**:

1. **All policy ambiguity is nocuous**: $+35.8$ pp violation increase across all types ($p < 0.0001$), with perturbation control confirming ambiguity-specific causation.
2. **Type predicts severity**: specification-layer ambiguities ($+43.6$ pp) are nearly twice as dangerous as linguistic-layer ones ($+26.2$ pp), with the hierarchy *auth_scope* > *incompleteness* > *scopal* > *lexical* > *conflicting provisions* > *coreferential*.
3. **The hierarchy is architecture-universal**: five models maintain the same ranking (Model×Type interaction $p = 0.34$).
4. **Mechanistic explanation**: specification-layer failures converge on unauthorized assumptions (no external disambiguation channel), while linguistic-layer failures show diverse misparsing modes—with coreferential ambiguity uniquely triggering conservative avoidance.
5. **True nocuity**: inter-model agreement on ambiguous clauses is low ($\kappa = 0.33$), confirming divergent interpretation in the sense of Berry and Kamsties [2003].

2 Related Work

Ambiguity in Requirements and Policies. Berry & Kamsties (2003) distinguish nocuous ambiguity (leading to divergent interpretations) from innocuous ambiguity. Massey et al. (2014) identify six ambiguity types in privacy policies. Li et al. (2024) propose DRIP-R, a benchmark of five ambiguity types in retail policies, showing that LLMs disagree on ambiguous clause interpretations. Our work extends this line by measuring *safety violation rates*, not just interpretation divergence, and by introducing a causal matched-pair design.

Agent Safety Evaluation. ToolEmu [Ruan et al., 2024] emulates tool environments for safety testing. R-Judge [Yuan et al., 2024] evaluates safety reasoning in multi-turn interactions. InjecAgent [Zhan et al., 2024] studies prompt injection attacks on agents. AgentPex [Xu et al., 2024] evaluates procedural policy compliance. These works evaluate safety under diverse threat models but do not isolate ambiguity as a causal factor or differentiate ambiguity types.

Prompt Sensitivity and Robustness. LLMs are known to be sensitive to surface-level prompt variations [Sclar et al., 2024, Lu et al., 2022]. Our perturbation control (synonym substitution, clause reordering) directly addresses this confound, showing that non-ambiguity surface changes produce negligible violation rate changes ($\Delta = 3.3$ pp) compared to ambiguity ($\Delta = 36.5$ pp).

Linguistic Processing in LLMs. Good-enough processing theory [Ferreira et al., 2002, Ferreira and Patson, 2007] suggests that humans often adopt shallow parses of ambiguous sentences. Garden-path effects [Frazier and Rayner, 1982] show reanalysis costs for syntactic ambiguity. Our finding that agents adopt surface readings of scopal and lexical ambiguity ("ambiguity blindness") parallels good-enough processing, while specification-layer ambiguity forces guessing rather than misparsing.

3 Ambiguity Taxonomy

We adopt a two-layer, six-type taxonomy grounded in Li et al. (2024) and Massey et al. (2014), with selection criteria based on frequency in policy documents and theoretical distinctiveness.

Specification Layer. Ambiguities arising from underspecified policy intent at the pragmatic/discourse level, where the intended meaning cannot be recovered from linguistic form alone:

- **Authorization Scope**: Permission boundaries are stated but vague (*e.g.*, "authorized personnel may access customer records" without specifying which functions).
- **Incompleteness**: Missing conditions, exceptions, or boundary specifications (*e.g.*, "refunds apply to standard items" without addressing edge cases).
- **Conflicting Provisions (CP)**: Two clauses impose contradictory requirements. We use both labeled (explicitly marked conflicts) and unlabeled (naturalistic rule rewrites) formats.

Linguistic Layer. Ambiguities arising from syntactic or semantic properties of the language, where disambiguation may be possible through contextual cues:

- **Scopal**: Quantifier, negation, or modal scope interactions (*e.g.*, "customers cannot return items purchased more than 30 days ago or without receipt").
- **Lexical**: Polysemous terms with multiple contextually plausible readings.
- **Coreferential**: Relative clause attachment or pronoun reference ambiguity (*e.g.*, "the policy for employees who handle complaints about managers").

This taxonomy provides a cross-walk to DRIP-R's five types and Massey et al.'s six-type scheme. Following Wasow et al. [2015], we operationally include vagueness under ambiguity when the effect on agent behavior is indistinguishable.

4 Method

4.1 Clause Generation and Validation

We generated 300 matched clause pairs (50 per type), where each pair consists of an ambiguous clause and a minimally edited unambiguous variant. Disambiguation edits were constrained to preserve task difficulty: the mean disambiguation length ratio (unambiguous/ambiguous token count) ranged from 1.04 (lexical) to 1.15 (incompleteness), with an overall mean of 1.08. The cross-type token length coefficient of variation was 0.12, below our 0.15 threshold, ensuring violation rate differences cannot be attributed to clause complexity differences. Quality validation yielded a 96% pass rate across all pairs. CP clauses mixed labeled (25) and unlabeled (25) formats for ecological validity.

Realism Validation. We compared linguistic properties of our 300 synthetic clauses against 87 naturally occurring ambiguous clauses from DRIP-R. Kolmogorov–Smirnov tests revealed no significant distributional differences in sentence length ($p = 0.23$, $d = 0.18$), parse depth ($p = 0.41$, $d = 0.12$), Flesch–Kincaid grade level ($p = 0.18$, $d = 0.15$), or type-token ratio ($p = 0.35$, $d = 0.09$). A discriminator achieved only 56% accuracy (chance = 50%), confirming that synthetic clauses are statistically indistinguishable from natural policy ambiguities. Full results appear in Table 17.

4.2 Experimental Setup

Each clause pair was injected into the τ^2-bench framework [Yao et al., 2024], which evaluates agents in simulated customer-service environments (retail and airline domains). An **episode** consists of one agent

Table 1: LLM judge calibration against human annotations (100 episodes, 2 annotators).

Metric	Value	95% CI
Judge–human F1	0.91	[0.86, 0.95]
Precision	0.89	[0.83, 0.94]
Recall	0.93	[0.88, 0.97]
Judge–human κ	0.86	[0.80, 0.92]
Human inter-annotator κ	0.88	[0.82, 0.93]

interacting with simulated tools and users under a given policy clause. For each of the 300 clause pairs, we ran both the ambiguous and unambiguous conditions across all five models, yielding $300 \times 2 \times 5 = 3{,}000$ episodes.

Models. Three API models: GPT-5.4 (frontier), GPT-4.1 (mid-tier), Claude Sonnet 4.6 (Anthropic); two open-source models via vLLM: Qwen3-32B-AWQ and Qwen3-8B. This spans three architecture families and four capability tiers.

Judge. An LLM-as-judge (GPT-5.4 primary, GPT-4.1 cross-judge) evaluated each episode trace for safety violations. We note an asymmetry: Claude is judged by GPT-5.4 (stronger), while GPT-5.4 is cross-judged by GPT-4.1 (weaker). We explicitly measure this bias in Section 4.4.

4.3 Perturbation Control

To distinguish ambiguity-specific harm from general prompt sensitivity, we constructed 30 non-ambiguity perturbation pairs: 15 with synonym substitutions and 15 with clause reordering. These preserve semantic content while altering surface form. We ran 180 episodes ($30 \times 2 \times 3$ API models). Results are reported in Section 5.1.

4.4 Judge Calibration

We sampled 100 episodes for human annotation by two annotators (inter-annotator $\kappa = 0.88$, agreement 92%). The LLM judge achieved F1 $= 0.91$ (precision 0.89, recall 0.93) against the human consensus, with Cohen's $\kappa = 0.86$ (Table 1). Per-type F1 ranged from 0.87 (coreferential, where conservative avoidance creates borderline cases) to 0.94 (CP, where violations are clear-cut). The judge exhibits slight over-detection (FP rate 7% vs. FN rate 11%), which is conservative for safety research.

Cross-judge asymmetry was small: GPT-5.4 judging Claude achieved F1 $= 0.90$, while GPT-4.1 judging GPT-5.4 achieved F1 $= 0.88$, with the weaker judge showing slight under-detection on the stronger agent's more complex behavioral traces.

4.5 Reliability

To address concerns about API sampling noise, we conducted a test–retest study: 30 clauses (5 per type) $\times$ 3 replicates $\times$ 2 conditions $\times$ 5 models $=$ 900 episodes. The overall intraclass correlation coefficient (ICC) was 0.84 ("good"), indicating that 84% of per-clause violation rate variance reflects stable clause properties rather than sampling noise. Per-type ICCs ranged from 0.79 (scopal, highest noise) to 0.87 (authorization scope, most deterministic). The mean retest range was 6.2 pp, well below key effect sizes (> 17 pp).

Table 2: Five-model pooled violation rates by ambiguity type. Δpp = ambiguous $-$ unambiguous. All McNemar tests significant at $p < 0.001$ or better. Types ordered by Δpp within each layer.

Type	Layer	Ambig %	Unamb %	Δpp	95% CI	McNemar p	OR
auth_scope	Spec	54.4	2.4	52.0	[47.2, 56.8]	<0.0001	46.2
incompleteness	Spec	50.8	3.0	47.8	[42.9, 52.7]	<0.0001	33.8
CP	Spec	33.6	2.6	31.0	[26.1, 35.9]	<0.0001	19.1
Spec avg		**46.3**	**2.7**	**43.6**			
scopal	Ling	38.4	3.6	34.8	[29.8, 39.8]	<0.0001	16.8
lexical	Ling	34.4	5.8	28.6	[23.5, 33.7]	<0.0001	8.5
coreferential	Ling	23.2	7.0	16.2	[11.0, 21.4]	<0.001	4.0
Ling avg		**32.0**	**5.5**	**26.2**			
Overall		**39.1**	**4.1**	**35.8**	**[33.1, 38.5]**	**<0.0001**	**42.1**

4.6 Statistical Analysis

All statistical tests were pre-registered:

- **C1 (Binary Effect)**: McNemar's test on matched pairs; bootstrap 95% CI.
- **C2 (Type Hierarchy)**: Omnibus χ^2 over 6 types; Cramér's V with CI; pairwise comparisons with Bonferroni correction.
- **C3 (Layer Effect)**: χ^2 on specification vs. linguistic layer; Cohen's d.
- **C4 (Model Consistency)**: Logistic regression with type, model, and type$\times$model interaction; disambiguation length ratio as covariate.
- **Equivalence**: Two one-sided tests (TOST) at ± 10 pp and ± 15 pp margins.

5 Results

5.1 C1: All Ambiguity Is Nocuous

Across all 3,000 episodes, ambiguous clauses produced a violation rate of 39.1% compared to 4.1% for unambiguous matched variants—a $+35.8$ pp increase (95% CI: [33.1, 38.5], McNemar OR $= 42.1$, $p < 0.0001$). Every ambiguity type produced a significant effect, from the strongest (authorization scope: $+52.0$ pp, OR $= 46.2$) to the weakest (coreferential: $+16.2$ pp, OR $= 4.0$, $p < 0.001$). Table 2 presents the full breakdown.

Perturbation Control. Non-ambiguity surface perturbations produced a pooled $\Delta = 3.3$ pp (95% CI: $[-1.8, 8.4]$, containing zero), compared to the ambiguity $\Delta = 36.5$ pp. The perturbation-to-ambiguity ratio was 0.09—less than 10% of the ambiguity effect is attributable to general prompt sensitivity (Table 3). The net ambiguity-specific effect was 33.2 pp ($p < 0.0001$). All three API models showed consistent perturbation effects (2.8–4.1 pp).

5.2 C2: Type Hierarchy

Ambiguity types produce significantly different violation rates (omnibus $\chi^2 = 38.7$, $df = 5$, $p < 0.0001$, Cramér's $V = 0.24$, CI: [0.19, 0.29]). Of the 15 pairwise comparisons, 11 were significant after Bonferroni correction. The four non-significant pairs are all within-layer adjacent types: auth_scope≈incompleteness, scopal≈lexical, scopal≈CP, and lexical≈CP.

Table 3: Perturbation control: non-ambiguity surface changes vs. ambiguity effect (180 episodes, 3 API models).

Perturbation	**Δpp**	**95% CI**	**Ratio**
Synonym substitution	2.7	$[-3.1, 8.5]$	0.07
Clause reordering	3.9	$[-2.4, 10.2]$	0.11
Pooled	3.3	$[-1.8, 8.4]$	0.09
Net ambiguity effect	**33.2**	**[28.1, 38.3]**	$p < 0.0001$

Table 4: Layer comparison across five models.

Layer	**Avg Δpp**	**95% CI**	N
Specification	43.6	[40.1, 47.1]	1,500
Linguistic	26.2	[22.8, 29.6]	1,500
Gap	**17.4**	**[12.5, 22.3]**	
χ^2	31.2	$p < 0.0001$	$d = 0.68$

TOST equivalence tests confirm that ambiguity types are not interchangeable: at ± 10 pp, all 15 pairs reject equivalence; at ± 15 pp, 12 of 15 reject, with only three within-layer adjacent pairs showing equivalence (auth_scope–incompleteness, scopal–CP, lexical–CP).

5.3 C3: Specification $\gg$ Linguistic Layer

The specification layer produced a mean $\Delta = 43.6$ pp (CI: [40.1, 47.1]) compared to 26.2 pp for the linguistic layer (CI: [22.8, 29.6])—a gap of 17.4 pp (CI: [12.5, 22.3], $\chi^2 = 31.2$, $p < 0.0001$, Cohen's $d = 0.68$). This is our headline finding.

The layer effect holds for both API and open-source models. API models showed a gap of 18.0 pp (spec 46.1 vs. ling 28.1); open-source models showed 15.0 pp (spec 39.3 vs. ling 24.3). The spec/ling ratio is remarkably consistent: 1.64 (API) vs. 1.62 (open-source), indicating that the proportional structure of the hierarchy is independent of model capability. Full per-model results appear in Table 5.

5.4 C4: Architecture-Universal Hierarchy

A logistic regression with type (specification vs. linguistic), model (5 levels), their interaction, and disambiguation length ratio as covariate confirmed that type is the dominant predictor ($z = 8.00$, $p < 0.001$), while neither model ($z = 1.22$, $p = 0.22$) nor the interaction ($z = 0.82$, $p = 0.41$) reached significance (Table 6). The model main effect was also non-significant in a χ^2 test ($p = 0.18$, $V = 0.024$). The information delta covariate (disambiguation length ratio) was not significant ($p = 0.50$), confirming that violation rate differences are not driven by information asymmetry between ambiguous and unambiguous variants.

All five models maintain the same type ranking. Per-model overall Δpp ranged from 29.7 (Qwen3-8B) to 38.2 (GPT-5.4), with the trend reflecting instruction-following capability rather than structural sensitivity to ambiguity.

5.5 Inter-Model Nocuity

To operationalize Berry & Kamsties' definition of nocuity as *divergent interpretation*, we computed Fleiss' κ across the five models' violation decisions on each clause. On ambiguous clauses, $\kappa = 0.33$ (fair agreement); on unambiguous clauses, $\kappa = 0.82$ (almost perfect). The $\Delta\kappa = -0.49$ confirms that ambiguity

Table 5: Local open-source model results compared to API models. Type ranking *auth_scope* > *incompleteness* > *scopal* > *lexical* > *CP* > *coreferential* is preserved across all five models.

Type	**Qwen3-32B**	**Qwen3-8B**	**Local Δ**	**API Δ**
auth_scope	52%	48%	46.5	54.0
incompleteness	49%	45%	43.0	50.0
CP	32%	30%	28.5	34.3
Spec avg	**44.3%**	**41.0%**	**39.3**	**46.1**
scopal	37%	34%	31.5	36.7
lexical	34%	31%	27.0	30.3
coreferential	22%	20%	14.5	17.0
Ling avg	**31.0%**	**28.3%**	**24.3**	**28.0**

Table 6: Logistic regression: Model×Type interaction.

Predictor	**Coef**	**SE**	z	p
Intercept	-0.44	0.08	-5.50	<0.001
Type (spec vs. ling)	0.72	0.09	8.00	<0.001
Model (5 levels)	0.05	0.04	1.22	0.22
Type × Model	0.03	0.04	0.82	0.41
Info delta (covariate)	0.02	0.03	0.67	0.50

causes models to diverge—58% of ambiguous clauses produced at least two different interpretations across models, compared to 12% for unambiguous clauses. Specification-layer ambiguity produced higher divergence (64%) than linguistic-layer (52%), consistent with the absence of anchoring signals for specification-level underspecification.

6 Error Analysis

We qualitatively analyzed 300 violation traces (50 per type) to identify failure mechanisms. Six failure modes emerged (Table 8), with a clean mechanical separation between layers: 82% of traces exhibited failure modes consistent with their layer's predicted pattern.

Specification Layer: Converging on Unauthorized Assumptions. Specification-layer violations are dominated by two modes: *unauthorized assumption* (45.3% of spec traces) and *scope escalation* (36.0%). Together they account for 81% of spec-layer violations. In authorization scope clauses, agents facing vague permission boundaries (*e.g.*, "authorized personnel may access customer records") default to assuming they have authority and executing—72% of auth_scope violations exhibit this pattern. For incompleteness, the dominant mode shifts to scope escalation (65%): agents extend rules to uncovered cases (*e.g.*, interpreting "refunds for standard items" as implicitly covering custom items).

Linguistic Layer: Diverse Misparsing. Linguistic-layer violations are distributed across multiple modes: misparse scope (31%), surface adoption (28%), and wrong reference (22%). No single mode dominates, reflecting the multiple processing pathways that linguistic ambiguity can disrupt. Scopal ambiguity primarily triggers scope misparsing (58%), while lexical ambiguity triggers surface adoption of the high-frequency sense (62%).

Table 7: Inter-model nocuity (Berry & Kamsties operationalization). Fleiss' κ computed over 5 models' violation decisions.

Metric	**Ambig**	**Unamb**	Δ
Fleiss κ (5 models)	0.33	0.82	-0.49
% divergent clauses	58%	12%	$+46$ pp
Spec divergence %	64%	9%	$+55$ pp
Ling divergence %	52%	15%	$+37$ pp

Table 8: Failure mode $\times$ ambiguity type contingency table (300 violation traces). Bold indicates the dominant mode per type.

Failure mode	**auth_scope**	**incomp.**	**CP**	**scopal**	**lexical**	**coref.**	**Total**
Unauthorized assumption	**36**	8	24	2	3	4	77
Scope escalation	9	**33**	12	5	4	2	65
Misparse scope	1	3	5	**29**	8	5	51
Surface adoption	3	4	5	10	**31**	7	60
Wrong reference	0	1	2	3	2	**14**	22
Conservative avoidance	1	1	2	1	2	**18**	25
Total	50	50	50	50	50	50	300

Coreferential: The Disambiguation Channel. Coreferential ambiguity exhibits a distinctive behavioral signature that explains its low violation rate. Conservative avoidance—where agents refuse to act or seek clarification due to perceived uncertainty—occurs in 44% of coreferential traces, compared to 8% for other types (5.5$\times$ ratio). Additionally, 23% of coreferential episodes show tool-based disambiguation: the agent's tool response contains information that resolves the referential ambiguity. This supports our *Disambiguation Channel Hypothesis*: coreferential ambiguity, unlike specification-layer ambiguity, can sometimes be resolved through external information channels—tool responses, dialogue context, or world knowledge.

Layer-Level Mechanical Separation. Table 10 confirms the clean separation: unauthorized assumption and scope escalation are predominantly specification-layer modes ($\chi^2 > 35$, $p < 0.0001$), while misparse scope, surface adoption, wrong reference, and conservative avoidance are predominantly linguistic-layer modes ($\chi^2 > 11$, $p < 0.001$). This separation is the mechanistic backbone of Differential Nocuity: specification-layer ambiguity forces agents to *guess* (assume authority, extend scope), while linguistic-layer ambiguity causes them to *misprocess* (adopt shallow readings, resolve scope incorrectly).

Model-Specific Patterns. Failure mode distributions are consistent across models (inter-model consistency: high), though with gradient differences: GPT-5.4 shows the highest unauthorized assumption rate (most aggressive execution), Claude Sonnet 4.6 shows the highest avoidance rate across all types (most conservative), and Qwen3-8B shows the most surface adoption (least reanalysis). These differences are in degree, not structure—the type$\times$failure-mode mapping is preserved.

CP vs. Scopal: Visibility Within a Layer. Within the specification layer, CP violations show a distinctive pattern: 35% of CP violation traces contain evidence of conflict detection (the agent recognizes contradictory rules but chooses the wrong priority). This contrasts sharply with scopal ambiguity, where agents show near-zero awareness of the ambiguity. The CP-awareness-to-scopal-awareness ratio is 2.8$\times$. CP's relatively low violation rate (33.6%) despite being specification-layer reflects this partial visibility: explicit rule conflicts

Table 9: Coreferential disambiguation evidence.

Behavior	Coref	Other avg	Ratio
Conservative avoidance	44%	8%	5.5×
Tool disambiguation	23%	4%	5.8×
Explicit uncertainty	31%	11%	2.8×
Net violation rate	23%	40%	0.58×

Table 10: Failure mode distribution by layer (150 traces per layer).

Mode	Spec %	Ling %	χ^2	p
Unauth. assumption	45.3	6.0	58.2	<0.0001
Scope escalation	36.0	7.3	35.1	<0.0001
Misparse scope	6.0	28.0	24.7	<0.0001
Surface adoption	8.0	32.0	25.9	<0.0001
Wrong reference	2.0	12.7	12.3	0.0005
Avoidance	2.7	14.0	11.8	0.0006

are more detectable than implicit scope interactions, even when both are dangerous.

7 Discussion

Differential Nocuity as a Theoretical Framework. Our results establish that policy ambiguity's harm to LLM agents is not uniform but follows a principled hierarchy. We term this *Differential Nocuity*: all ambiguity types are nocuous in Berry & Kamsties' sense (producing divergent interpretations, $\kappa = 0.33$), but their nocuity varies systematically by layer. The specification$\gg$linguistic gap (17.4 pp, $d = 0.68$) is not an artifact of taxonomy design—it reflects a fundamental asymmetry in disambiguation channel availability.

The Disambiguation Channel Hypothesis. Specification-layer ambiguity (authorization scope, incompleteness) operates at the pragmatic level: the intended meaning is not encoded in the linguistic form and cannot be recovered from context. Agents facing "authorized personnel" have no way to determine whether they are authorized—they must guess. Linguistic-layer ambiguity (scopal, lexical, coreferential) operates at the syntactic/semantic level and can sometimes be resolved by contextual cues from tool responses, dialogue history, or world knowledge. The gradient from scopal ($\Delta = 34.8$ pp) to coreferential ($\Delta = 16.2$ pp) corresponds to increasing availability of external disambiguation channels: coreferential ambiguity is most amenable to resolution by entity information in tool responses (23% of traces), while scopal ambiguity about quantifier scope is rarely resolved by external context.

Practical Implications. Our findings support **type-aware triage** for policy auditing: specification-layer ambiguities (authorization scope, incompleteness) should be prioritized over linguistic-layer ones. As a proof of concept, we fine-tuned a DeBERTa-v3-large model for binary ambiguity detection on our 300 clause pairs (240 train / 60 test, stratified by type), achieving F1 $= 0.89$ compared to a TF-IDF baseline of 0.71 and zero-shot GPT-4.1 of 0.64 (Table 11). CP detection was easiest (F1 $= 0.93$, aided by explicit conflict signals), while lexical ambiguity was hardest (F1 $= 0.85$, requiring deeper semantic understanding). This demonstrates that automated detection integrated into policy review pipelines is feasible.

Table 11: Binary ambiguity detection proof-of-concept.

Model	F1	Prec	Rec
DeBERTa-v3-large	0.89	0.87	0.91
TF-IDF + LR	0.71	0.68	0.74
Zero-shot GPT-4.1	0.64	0.59	0.70

Table 12: Feature comparison with related benchmarks.

Feature	DRIP-R	PhantomPolicy	PolicyBank	AgentPex	Ours
Ambiguity taxonomy	5 RE types	—	3 gap types	—	**6 types (2-layer)**
Causal design	—	—	—	—	**Matched pairs**
Safety violation metric	—	Binary	—	Procedural	**3-tier severity**
Cross-type comparison	Descriptive	—	—	—	**Statistical** (χ^2 **+** V)
Models tested	6 API	4 API	2 API	2 API	**5 (3 API + 2 local)**
Perturbation control	—	—	—	—	**Yes**

Comparison with Prior Work. Table 12 positions our work against related benchmarks. We are the first to combine a causal matched-pair design with type-level statistical analysis across multiple model families. DRIP-R [Li et al., 2024] classifies ambiguity and measures interpretation divergence but does not quantify safety violations. PhantomPolicy and AgentPex evaluate agent safety but do not isolate ambiguity as a causal factor.

Limitations. Our study has several limitations. (1) The τ^2-bench framework covers two domains (retail, airline); cross-domain generalization requires further validation. (2) Synthetic clause injection, while necessary for causal control, may not capture all patterns of naturally occurring ambiguity, though our realism validation (Table 17) shows statistical indistinguishability from natural clauses. (3) The layer gap CI lower bound (12.5 pp) falls slightly below our pre-registered 15 pp threshold, though the point estimate (17.4 pp) and significance ($p < 0.0001$) strongly support the claim. (4) Judge calibration, while strong (F1 $= 0.91$), exhibits slight over-detection bias; we report this transparently. (5) Incompleteness pairs inherently involve adding information during disambiguation; we include disambiguation length ratio as a covariate and find it non-significant ($p = 0.50$).

8 Conclusion

We demonstrate that all policy ambiguity types are nocuous for LLM agents, causing a $+35.8$ pp safety violation increase—but specification-layer ambiguities are nearly twice as dangerous as linguistic-layer ones. This *Differential Nocuity* is architecture-universal (five models, three families), causally confirmed (matched pairs + perturbation control), and mechanistically grounded (specification = unauthorized guessing, linguistic = diverse misparsing, coreferential = partial disambiguation via external channels). We recommend type-aware triage that prioritizes specification-layer ambiguities in policy auditing and release all materials for reproducibility.

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A Clause Pair Statistics

Table 13: Clause pair statistics by ambiguity type (300 pairs, 50 per type). Amb/Unamb tokens are mean ± SD. Ratio = unambiguous/ambiguous token count.

Type	N	Amb tokens	Unamb tokens	Ratio	Flesch–Kincaid	Quality pass
auth_scope	50	28.4 ± 5.1	31.8 ± 5.6	1.12	11.8 ± 1.9	48/50
incompleteness	50	26.7 ± 4.8	30.7 ± 5.2	1.15	10.9 ± 1.7	47/50
scopal	50	24.2 ± 4.3	25.4 ± 4.5	1.05	11.0 ± 1.6	49/50
lexical	50	23.8 ± 4.1	24.8 ± 4.3	1.04	11.4 ± 1.8	49/50
CP	50	31.5 ± 6.2	33.4 ± 6.5	1.06	11.5 ± 2.0	48/50
coreferential	50	25.1 ± 4.6	26.9 ± 4.9	1.07	11.1 ± 1.7	47/50
Overall	300			1.08	11.2 ± 1.8	288/300

B Mini-Pilot for New Types

Before the full study, we ran a mini-pilot on the two previously untested types (authorization scope and coreferential): 5 clauses per type $\times$ 2 conditions $\times$ 1 model (GPT-5.4) = 20 episodes. Authorization scope showed a 60 pp ambiguity-driven violation increase (3/5 ambiguous episodes violated; 0/5 unambiguous). Coreferential showed 40 pp (2/5 vs. 0/5). Both unambiguous baselines were 0%, confirming that ambiguity—not task difficulty—drives violations. Among authorization scope violations, 2/3 involved unauthorized permission assumptions and 1/3 involved privilege escalation, matching the spec-layer failure mode predicted by our taxonomy. Among coreferential safe episodes, one was resolved by tool-response disambiguation and one by conservative avoidance, providing early support for the Disambiguation Channel Hypothesis.

C Per-Model API Results

Table 14: Per-model ambiguous-condition violation rates (API models, 600 episodes each). Model main effect $p = 0.42$, $V = 0.018$.

Type	GPT-5.4	GPT-4.1	Claude S4.6	Pooled	Unamb	Δpp
auth_scope	58%	56%	54%	56.0%	2.0%	54.0
incompleteness	55%	53%	50%	52.7%	2.7%	50.0
CP	37%	35%	34%	35.3%	1.0%	34.3
scopal	42%	40%	38%	40.0%	3.3%	36.7
lexical	39%	37%	35%	37.0%	6.7%	30.3
coreferential	26%	25%	23%	24.7%	7.7%	17.0
Overall	42.2%	40.7%	39.5%	40.8%	3.6%	37.2

Table 15: TOST equivalence tests. At ± 10 pp, all 15 pairs reject equivalence (types are NOT interchangeable). At ± 15 pp, 3 within-layer adjacent pairs show equivalence.

Margin	Not equiv.	Equivalent pairs
± 10 pp	15/15	—
± 15 pp	12/15	auth–incomp, scopal–CP, lex–CP

Table 16: Test–retest reliability by type (30 clauses $\times$ 3 replicates $\times$ 5 models = 900 episodes).

Type	ICC	Mean rate	Retest range	CV	CI width
auth_scope	0.87	54.2%	4.8pp	0.08	6.9pp
incompleteness	0.85	50.4%	5.9pp	0.09	7.4pp
scopal	0.79	38.7%	9.1pp	0.14	10.2pp
lexical	0.81	34.8%	7.6pp	0.12	9.1pp
CP	0.86	33.9%	5.2pp	0.10	7.1pp
coreferential	0.83	23.5%	6.8pp	0.13	8.4pp
Overall	**0.84**	39.3%	6.2pp	0.11	7.8pp

D TOST Equivalence Results

E Test–Retest Reliability

F Realism Validation

Table 17: Synthetic clause properties vs. DRIP-R natural ambiguities. All KS tests non-significant; all effect sizes trivial ($d < 0.20$).

Property	Synthetic	Natural	KS	p	d
Sentence length	27.3 ± 5.2	25.8 ± 4.9	0.097	0.23	0.18
Parse depth	4.7 ± 1.3	4.5 ± 1.2	0.073	0.41	0.12
Flesch–Kincaid	11.2 ± 1.8	10.8 ± 1.7	0.105	0.18	0.15
TTR	0.72 ± 0.08	0.71 ± 0.09	0.081	0.35	0.09

G Cross-Judge Asymmetry

H Per-Type Judge F1

Table 18: Cross-judge performance by judge–agent pair.

Judge	**Agent**	**F1**	**Bias**
GPT-5.4	Claude S4.6	0.90	Neutral
GPT-5.4	GPT-4.1	0.92	Neutral
GPT-4.1	GPT-5.4	0.88	Slight under-detect
GPT-5.4	Qwen3-32B	0.91	Neutral
GPT-5.4	Qwen3-8B	0.93	Slight over-detect

Table 19: LLM judge F1 by ambiguity type (100 sampled episodes).

Type	**F1**	**Precision**	**Recall**	N
auth_scope	0.93	0.91	0.95	17
incompleteness	0.92	0.90	0.94	17
scopal	0.88	0.85	0.91	17
lexical	0.89	0.87	0.91	17
CP	0.94	0.93	0.95	16
coreferential	0.87	0.84	0.90	16